INCH-POUND

ZZ-H-561K
July 26, 1988
SUPERSEDING
ZZ-H-561J
September 16, 1983

FEDERAL SPECIFICATION

HOSE, RUBBER, AND HOSE ASSEMBLIES, RUBBER, SMOOTH BORE, WATER SUCTION AND DISCHARGE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wire-reinforced hose and hose assemblies for use with water pumps. The hose and hose assemblies shall be usable on both suction and discharge sides of the pump.

1.2 Classification.

1.2.1 Grades, styles, and classes. Water hose shall be of the following grades, styles, and classes as specified (see 6.2.1).

Grade A - Weather-, oil-, and ozone-resistant outer cover
Grade B - Weather-, and ozone-resistant outer cover
Style A - Hose ends constructed for shank, compression, or internally expanded fittings.
Style C - Hose ends with built-in nipples.
Style D - Hose ends constructed for straight nipples
Class 1 - General water use inner tube
Class 2 - Potable water use inner tube

1.2.2 Sizes. Water hose shall be of the following sizes as specified (see 6.2.1). 1.0, 1.25, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 6.0, 8.0, 10.0, and 12.0 inches inside diameter.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications

WW-C-440 - Clamps, Hose, (Low-Pressure)
WW-C-624 - Coupling Assembly, Hose (Garden, Water, and Water Suction)

Federal Standards

FED-STD-123 - Marking for Shipment (Civil Agencies)
FED-STD-162 - Hose, Rubber, Visual Inspection Guide for

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal activities may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions, and the Index of Federal Specifications and Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric or Metal (including Tubing) and Fittings, Nozzles and Strainers)
MIL-C-38404 - Couplings, Hose, Reattachable Screw-on

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Federal Regulation

21 CFR 177 - Federal Food and Drugs Act; Indirect Food Additives

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

A53 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated Welded and Seamless
A153 - Zinc Coating (Hot-Dip) on Iron and Steel Hardware
D297 - Rubber Products-Chemical Analysis
D380 - Rubber Hose
D412 - Rubber Properties in Tension
D413 - Rubber Property-Adhesion to Flexible Substratum
D471 - Rubber Property-Effect of Liquids
D518 - Rubber Deterioration-Surface Cracking
D573 - Rubber Deterioration in an Air Oven
D1149 - Rubber Deterioration-Surface Ozone Cracking in a Chamber (Flat Specimen)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1, 6.2.1, and 6.3).

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3.2 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Design and construction. The hose shall consist of an inner tube of elastomeric material, layer or layers of textile reinforcement, a helical wire, reinforcement, additional layer or layers of textile reinforcement, or the textile layer may be interwoven with the helical wire, and an outer cover of elestomeric material.

3.4.1 Outer cover. The outer cover shall be grade A or grade B as specified (see 6.2.1).

3.4.1.1 Grade A. The grade A outer cover shall be constructed of an elastomeric material that is weather-, oil-, and ozone-resistant.

3.4.1.2 Grade B. The grade B outer cover shall be constructed of an elastomeric material that is weather and ozone-resistant.

3.4.2 Inner tube. The inner tube shall be class 1 or class 2 as specified (see 6.2.1).

3.4.2.1 Class 1. The class 1 inner tube shall be constructed of an elastomeric material that is suitable for general water use.

3.4.2.2 Class 2. The class 2 inner tube shall be constructed of an elastomeric material that meets the requirements of the Federal Food and Drugs Act, Title 21, Code of Federal Regulations 177.2600. It shall meet the extraction requirements of 3.6.3 when tested as specified in 4.5.7.

3.4.3 Reinforcement.

3.4.3.1 Textile reinforcement. Textile reinforcement shall consist of one or more plies of loomed, woven fabric, braided or spiraled yarn of cotton or synthetic material, evenly and firmly applied. It shall be free of unsightly

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defects, dirt, knots, lumps, and irregularities of twist. It shall be applied above and below the helix of wire reinforcement and shall be well impregnated with an elastomeric material.

3.4.3.2 Wire reinforcement. A helix of wire, completely embedded in an elastomeric material, shall be installed between the upper and lower ply or plies of textile reinforcement, or the helix of wire may be interwoven with the textile reinforcement. The reinforcing wire shall be round, galvanized, coppered, or unfinished (bright) steel wire. The size, tensile strength, and spacing of the wire reinforcement shall be sufficient to meet the requirements specified herein. Corrugation in the hose shall not exceed that consistent with good manufacturing practice. When tested in accordance with 4.5.13 the wire shall show no defects in the steel.

3.4.4 Length. Hoses and hose assemblies shall be furnished in lengths as specified (see 6.2.1) with a tolerance of ±2 percent when tested in accordance with 4.5.1 (see 6.4). Nipples, fittings, and couplings shall be considered additions to hose length, not part of the length.

3.4.5 Hose ends. Unless otherwise specified, hose ends shall be of a style that is appropriate for the type of fitting or coupling specified (see 3.5). When specified (see 6.2.1), hose ends shall be style A, C, or D.

3.4.5.1 Style A. When style A ends are specified, the helix reinforcement wire shall be terminated at the end of the hose (see figure 1A).

3.4.5.2 Style C. When style C ends are specified (see 6.2.1), each end of style C hose (4 inches in diameter and larger) shall be fitted with nipples conforming to size and dimension of figure 5. Nipples shall be made of steel pipe conforming to ASTM A53; schedule 40 for 4 inch and 6 inch sizes and schedule 30 for 8 inch, and 12 inch sizes. Bands on the nipples shall be steel and welded all the way around the pipe. The nipple shall be zinc-coated in accordance with ASTM A53 or otherwise adequately protected at exposed ends from corrosion after machining. The helix reinforcement wire shall extend over the end of the built-in nipple at least 1.5 inches under or over the wire binding the nipples (see figure 1B). At least one ply of fabric shall be between the binding wire and the cover. The nipple shall be secured to the hose by the vulcanization process. The nipple shall extend beyond the end of the hose to the length specified in table 1 (see figure 1).

3.4.5.3 Style D. When specified (see 6.2.1), each end of style D hose (4 inches in diameter and larger) shall be fitted with clamps and with nipples conforming to figure 6. Nipples shall be made of steel pipe conforming to ASTM A53. The nipple shall be zinc-coated in accordance with ASTM A53 or otherwise adequately protected at exposed ends from corrosion after machining. The inside diameter of the hose ends shall be large enough to accommodate the outside diameter of schedule 80 iron pipe.

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TABLE I. Design and performance requirements.

Requirements	Nominal size in inches (inside diameter)												
	1.0	1.25	1.5	2.0	2.5	3.0	3.5	4.0	4.5	6.0	8.0	10.0	12.0
Inside diameter, inches	1.0	1.25	1.5	2.0	2.5	3.0	3.5	4.0	4.5	6.0	8.0	10.0	12.0
Tolerance: Plus	0.031	0.031	0.031	0.063	0.063	0.063	0.063	0.063	0.063	0.063	0.063	0.063	0.063
Minus	0.031	0.031	0.031	0.063	0.063	0.063	0.063	0.250	0.250	0.250	0.250	0.250	0.250
Thickness, inch, minimum1/													
Tube	0.047	0.047	0.047	0.047	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
Cover	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.063	0.063	0.063	0.063
Adhesion, Pounds per inch of width, minimum:													
Between plies and rubber parts	12	12	12	12	12	12	12	12	12	12	12	12	12
Between plies	12	12	12	12	12	12	12	12	12	12	12	12	12
Tensile strength (before aging), Pounds per square (lb/in^2) minimum:													
Tube	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Cover	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Ultimate elongation, minimum percent:													
Tube	200	200	200	200	200	200	200	200	200	200	200	200	200
Cover	200	200	200	200	200	200	200	200	200	200	200	200	200
Hydrostatic tests lb/in^2													
Proof pressure 2/	100	100	100	100	100	100	100	100	100	100	100	100	100
Burst pressure	300	300	300	300	300	300	300	300	300	300	300	300	300
Crush resistance (pounds)	200	200	225	250	325	325	325	325	325	325	325	325	400
Length of nipples beyond hose end, inches, minimum:	--	--	--	--	--	--	--	4	4	6	8	8	8

1/ Minimum thickness of tube and cover before buffing.
2/ The hose, complete with couplings or nipples, shall withstand the pressure indicated.

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3.5 Fittings and couplings. Hose and hose assemblies shall be fitted with fittings and couplings as specified (see 6.2.1). Unless otherwise specified (see 6.2.1), each length of hose, up to and including 4 inches inside diameter, shall be fitted with ribshank or expansion couplings (see 3.5.3) or compression-reattachable couplings (see 3.5.4). Female couplings shall be fitted with a synthetic rubber washer.

3.5.1 Flanges. Unless otherwise specified (see 6.2.1), each nipple shall be fitted with a standard steel flange for 150 pound (lbs) working pressure. All steel surfaces shall be zinc-coated in accordance with ASTM D153.

3.5.1.1 Flanges for Navy requirements: When specified (see 6.2.1), flanges shall be in accordance with the requirements of figures 2, 3, and 4.

3.5.2 Clamps. Clamps shall be furnished and fitted when shank couplings or nipples without built-up bands are specified. Clamps conforming to WW-C-440 shall be corrosion-resistant steel, zinc-coated steel in accordance with ASTM A153, or copper alloy as specified (see 6.2.1). The clamps shall be capable of holding nipples and couplings in place without leakage or slippage when subjected to the proof pressure specified in table I.

3.5.3 Shank and expansion couplings. Unless otherwise specified (see 6.2.1), shank and expansion couplings shall conform to WW-C-624.

3.5.4 Compression-reattachable couplings. Unless otherwise specified (see 6.2.1), compression-reattachable couplings shall conform to MIL-C-38404.

3.6 Performance. Hose and hose assemblies shall meet the mechanical and physical properties of table I and the requirements of 3.6.1 through 3.6.6.

3.6.1 Hydrostatic pressure requirements. Hose and hose assemblies shall be subjected to the hydrostatic pressure specified in table I.

3.6.1.1 Proof pressure. When tested in accordance with 4.5.3, at the proof pressure specified in table I, the hose or hose assemblies shall not leak or show any imperfections in either the hose or the fittings. There shall be no indications of slippage of pullout of the fitting from the hose.

3.6.1.2 Burst pressure. When tested in accordance with 4.5.4, there shall be no failure of the hose or hose assembly at the minimum burst pressure specified in table I.

3.6.2 Crush resistance. During the crush resistance test specified in 4.5.5, the smallest outside diameter shall be not less than 85 percent of the original outside diameter. After release of the load, the smallest outside diameter shall be not less than 95 percent of the original outside diameter.

3.6.3 Water extraction of class 2 inner tube. The extractable nonvolatile matter in class 2 inner tube shall not exceed 21 milligram per square inch when the tubes are tested as specified in 4.5.7.

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3.6.4 Oil resistance of grade A hose cover. The tensile strength of the cover stock of grade A hose shall not decrease more than 60 percent of the original value after being subjected to ASTM No. 3 oil conforming to ASTM D471 in accordance with 4.5.9. In addition, the maximum permissible volume swell in ASTM No. 3 oil shall be 100 percent.

3.6.5 Accelerated aging. The tensile strength of the tube and cover after the accelerated aging process specified in 4.5.10 shall not decrease by more than 20 percent of the original value while the elongation of the tube and cover after similar aging shall not decrease by more than 50 percent.

3.6.6 Ozone resistance. The cover stock of grade A or grade B hose at the end of the exposure time (see 4.5.11) shall show no visible cracking under 7 times magnification.

3.7 Strainers and spanner wrench. When specified (see 6.2.1), suction strainers and spanner wrenches shall be furnished.

3.8 Identification marking.

3.8.1 General water use hose (class 1). Each length of hose shall be marked in a color that contrasts with the color of the hose cover. Marking shall be accomplished either by in laying a rubber or suitable material brand or by applying a suitable composition ink bonding the marking onto the cover in such a manner that the marking ink cannot be removed except by mechanical means. The marking shall consist of the manufacturer's name or trademark, the quarter and year of manufacture, the words Water - Suction, the symbol ZZ-H-561, the grade, the style, the class of the hose, and the words NOT FOR USE WITH POTABLE WATER. Hose shall be marked at regular intervals not exceeding 25 feet. Letters shall be at least 0.25-inch high. An alternative method of marking may be by application of a continuous embossed strip along the entire length, vulcanizing the hose, and subsequently removing the strip, leaving a continuous relief identification area. When marking is accomplished by this alternative, no color contrast is required.

3.8.2 Potable water use hose (class 2). When class 2 hose is specified (see 6.2.1), each hose length shall be marked in the same manner as specified in 3.8.1), except that the words POTABLE WATER USE ONLY shall be substituted for NOT FOR USE WITH POTABLE WATER.

3.9 Workmanship. The finished hose shall be uniform in thickness. The inner tube and the cover shall be smooth and free from pits and blisters. Reinforcement wire shall not be crushed and shall not protrude through the cover or tube.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on hose or hose assembly when a first article is required (see 3.1 6.2.1 and 6.3). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract. In addition to the first article, the following shall be submitted:

a. A 3-foot length of hose with couplings for the burst test (see 4.5.4).
b. A 1-foot length of hose for the crush test (see 4.5.5).
c. A 4-foot length of hose without helix wire for the physical tests (see 4.5.6 through 4.5.12).
d. A 6-foot length of helix wire for the flexibility test (see 4.5.13).

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

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4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one hose length or hose assembly. All hose and hose assemblies of the same grade, style, class, and size offered for delivery at one time shall be offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.2 Sampling for tests.

4.3.2.1 Sampling for length and inside diameter tests. Tests specified in 4.5.1 and 4.5.2 shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

4.3.2.2 Sampling for proof pressure test. All hose submitted shall be subjected to the hydrostatic proof pressure test specified in 4.5.3.

4.3.2.3 Sample for burst pressure test. A 3-foot length of hose with couplings identical to the hose submitted for the lot shall be submitted for the burst pressure test specified in 4.5.4.

4.3.2.4 Sample for crush resistance test. A 1-foot length of hose identical to the hose submitted for the lot shall be submitted for the crush resistance test specified in 4.5.5.

4.3.2.5 Sample for physical tests. A 4-foot length of hose identical to the hose submitted for the lot, except that it shall be without helix wire, shall be submitted for the physical tests specified in 4.5.6 through 4.5.12.

4.3.2.6 Sampling for helix wire test. A 6-foot length of helix wire taken from the same lot of wire used in manufacture of the hose shall be submitted for the helix wire flexibility test specified in 4.5.13.

4.4 Examination. Each of the sample hose lengths selected in accordance with 4.3 shall be examined for compliance with the requirements in section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements. The classification of defects in FED-STD-162 shall be used to determine and evaluate defects through visual inspection. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.5 Tests. Each hose length selected shall be subjected to the applicable tests specified in table II and 4.5.1 through 4.5.13. Failure to comply with the requirements of section 3 shall constitute cause for rejection.

4.5.1 Hose length measurement. Each hose selected shall be measured for length in accordance with ASTM D380. Prior to measurement of length, the hose shall be elongated by 10 lb/in2 water pressure for 10 minutes (see 6.4).

4.5 2 Inside diameter. Hose selected shall have the inside diameter measured in accordance with ASTM D380.

TABLE II. Test method.

Test	Method	Reference
Hose size, inside diameter	ASTM D380	table 1 and 4.5.2
Hose length	ASTM D380	3.4.4 and 4.5.1
Hydrostatic tests:		
Proof pressure	ASTM D380	3.6.1.1 and 4.5.3
Burst pressure	ASTM D380	3.6.1.2 and 4.5.4
Crush resistance		table 1, 3.6.2 and 4.5.5
Water extraction		3.6.3 and 4.5.7
Resistance to oil	ASTM D471	3.6.4 and 4.5.9
Accelerated aging	ASTM D573	3.6.5 and 4.5.10
Ozone resistance	ASTM D518 and D1149	3.6.6 and 4.5.11
Tube and cover thickness	ASTM D380	table 1 and 4.5.6
Friction (adhesion)	ASTM D380 and D413	table 1 and 4.5.12
Tensile strength and ultimate elongation of tube and cover	ASTM D412	table 1 and 4.5.8
Helix wire flexibility		4.5.13

4.5.3 Proof pressure test. Each hose length complete with couplings shall be subjected to the applicable hydrostatic test pressure specified in table I in accordance with ASTM D380, except that the pressure shall be maintained for 2 minutes. Any blistering or leaking of the hose shall be cause for rejection.

4.5.4 Burst pressure test. Each hose or hose assembly selected shall be subjected to the applicable hydrostatic test pressure specified in table I in accordance with ASTM D380.

4.5.5 Crush resistance test. Crush resistance shall be determined by centering a 12-inch length of hose between 3-inch wide, parallel metal plates in such a way that a 3-inch length of hose is being compressed. The plates shall be brought together at a rate of 2 inches per minute until the

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applicable crush resistance load specified in table I has been applied. The distance between plates shall be measured with a steel rule and expressed in percent of the original outside diameter. The load shall be released and the minimum outside diameter of the hose at the center of the compressed area shall be measured and reported in percent of the original outside diameter.

4.5.6 Tube and cover thicknesses. The thicknesses of the tube and cover shall be measured on specimens removed from each end of the hose section obtained in accordance with 4.3.2.5 to determine conformance to 3.6. The thicknesses shall be determined by the procedure given in ASTM D380. The respective values found for the tube thickness shall not be averaged, but shall be regarded as independent measurements, each of which should meet the requirement. The respective values for the cover thicknesses shall be treated in the same manner.

4.5.7 Water Extraction of class 2 hose test. The inner tube surface shall be extracted in distilled water at reflux temperature in accordance with the following procedure:

a. Remove a 1-inch ring of rubber inner tube and cut or buff away the outer surface until the surface is smooth.
b. Cut sufficient length of inner tube to obtain approximately 10 grams (g), based on a minimum thickness of 0.047 inch.
c. Subject sample to distilled water, at reflux temperature, for 7 hours in an extraction apparatus as shown in ASTM D297, figure 1.
d. Filter the solution through No. 40 Whatman filter paper or equivalent and collect the extract in a tared container.
e. Extract the rubber sample in an additional 50-75 cubic centimeters (cm^3) of distilled water at a reflux temperature for 2 more hours and repeat step d. Combine the extract solutions and evaporate to dryness. The container and residue shall be dried in an oven at 221° ± 5° degrees Fahrenheit (°F) for 1 hour, cooled in a desiccator and weighed. The weight of the residue shall be recorded. A blank shall be run using the same amount of distilled water. After making allowance for the blank, the weight of the residue shall be recorded to the nearest milligram.
f. Report the amount of matter extracted from the rubber in milligrams per square inch of surface.

4.5.8 Initial tensile strength and ultimate elongation test. The tensile strength and ultimate elongation of the cover and tube of the hose shall be determined in accordance with ASTM D412.

4.5.9 Oil resistance test of grade A hose cover. The rubber cover of grade A hose selected shall be subjected to ASTM oil No. 3 for 70 hours at 212°F ± 3.6°F in accordance with the method specified in ASTM D471. The change in tensile strength and amount of volume swell shall be based on original cross-sectional areas.

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4.5.10 Accelerated aging test. The resistance of the tube and cover to accelerated aging shall be determined as described in ASTM D573 except that the time of aging shall be 70 hours and the temperature of aging shall be 212°F. Tensile strength and ultimate elongation tests shall be used to determine the amount of deterioration.

4.5.11 Resistance to ozone test. The sample of grade A or grade B cover stock shall be tested for ozone resistance in accordance with procedure B of ASTM D518 and ASTM D1149 except that after conditioning for 24 hours in an ozone-free atmosphere the looped sample shall be exposed for 72 hours at 104°F ± 3.6°F to an atmosphere containing 50 parts per hundred million of ozone.

4.5.12 Adhesion test. From hose selected, prepare ring or strip specimens as described in ASTM D380. The adhesion shall be determined in accordance with the machine method of ASTM D413.

4.5.13 Helix wire flexibility test. The helix wire shall be wrapped six full tight turns around a mandrel having a diameter three times the diameter of the wire. The wrapped wire shall show no defects in the steel after wrapping. Cracking or breaking of the wire shall constitute a defect. Flaking or peeling of the galvanized or coppered coating of the wire shall not be considered a defect (see 3.4.3.2).

4.6 Preparation for delivery inspection. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one unit prepared for shipment. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with the requirements of MIL-P-775 with the level of preservation and packaging the level of packing as specified (see 6.2.1).

5.2 Marking.

5.2.1 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

5.2.2 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Intended use.

6.1.1 Class 1 hose. Class 1 hose and hose assemblies are intended for mines, construction work, general contracting, and general suction service.

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6.1.2 Class 2 hose. Class 2 hose and hose assemblies are intended as a means of conveying potable water in suction and discharge services, such as distillation units and tank farms.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Grade, style, and class required (see 1.2.1, 3.4.1, 3.4.2, 3.4.5 through 3.4.5.3 and 3.8.2).
c. Size required (see 1.2.2).
d. If first article is required for inspection and approval (see 3.1, 4.2.1, and 6.3).
e. Length required (see 3.4.4).
f. Type of fittings or couplings and clamps required (see 3.4.5.2, 3.4.5.3, and 3.5 through 3.5.4).
g. If other than 150 pound, standard steel flanges are required (see 3.5.1, and 3.5.1.1).
h. If clamps are to be corrosion-resistant steel, zinc-coated steel, or copper alloy (see 3.5.2).
i. If shank or expansion couplings are to be other than as specified (see 3.5.3).
j. If compression-reattachable couplings are to be other than as specified (see 3.5.4).
k. If suction strainers and spanner wrenches are required (see 3.7).
i. Level of preservation and level of packing required (see 5.1).

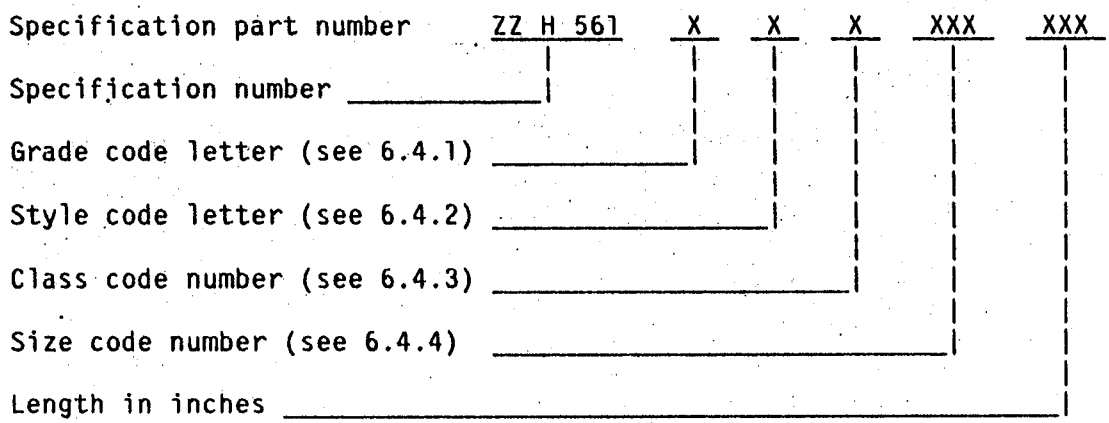
6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.3 First article. When a first article inspection is required (see 3.1 and 6.2.1), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of a sample hose or hose assembly. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Specification part number. The specification part number is a definitive part number which will be formulated to identify each item covered by this specification. The part number will be formulated by selecting from the requirement options available in this specification as follows:

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Hose assembly.



6.4.1 Hose grade. Hose grade is designated by one code letter (see table III).

TABLE III. Hose grade code letter.

Hose grade code letter	A	B
Hose grade	Grade A	Grade B

6.4.2 Hose style. Hose style is designated by one code letter (see table IV).

TABLE IV. Hose style code letter.

Hose style code letter	A	C	D
Hose style	Style A	Style C	Style D

6.4.3 Hose class. Hose class is designated by a one-digit code number (see table V).

TABLE V. Hose class code number.

Hose class code number	1	2
Hose class	Class 1	Class 2

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6.4.4 Hose size. Hose size is designated by a three-digit code number (see table VI).

TABLE VI. Hose size code number.

Hose size code number	010	012	015	020	025	030
Hose ID (inches)	1.0	1.25	1.5	2.0	2.5	3.0

Hose size code number	035	040	045	060	080	100	120
Hose ID (inches)	3.5	4.0	4.5	6.0	8.0	10.0	12.0

6.5 Hose length measurement. Hose containing an embedded helical wire tends to shrink abnormally during shipment and storage. Actual length should be determined by measurement of the hose while it is under 10 lb/in^2 pressure.

6.6 Cross-reference of classifications. The following hoses were previously classified as indicated:

ZZ-H-561H	ZZ-H-561G and ZZ-H-561H	ZZ-H-561J and ZZ-H-561K
Grade A	Grade A	Grade A
Grade B	Grade B	Grade B
Type I	Type I	Not designated
Type II	Type II	Not designated
Not designated	Style A	Style A
Not designated	Style B	Style A
Not designated	Style C	Style C
Not designated	Style D	Style D
Not designated	Class 1	Class 1
Not designated	Class 2	Class 2

6.7 Subject term (key word) listing.

Rubber
Hose
Water suction
Water discharge
Hose assemblies

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MILITARY INTEREST:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review Activities

Air Force - 82
DLA - CS

User Activities

Navy - MC, SH

CIVIL AGENCY COORDINATING ACTIVITIES:

DOT - ACO
GSA - FSS
HHS - FEC
USDA - AFS

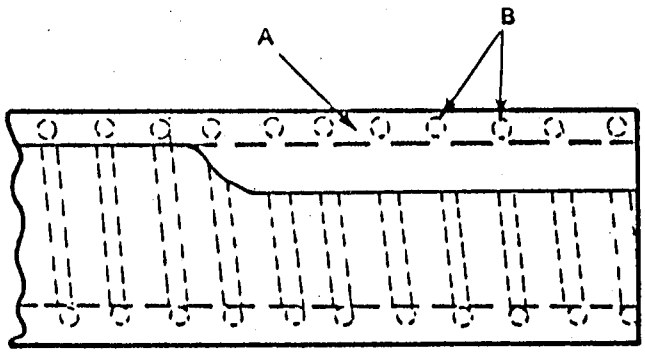
PREPARING ACTIVITY:

Navy - YD

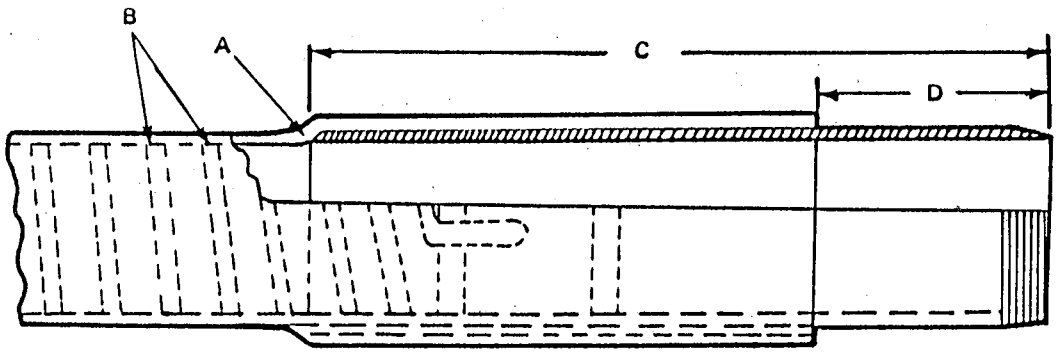
Project No. 4720-0760

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

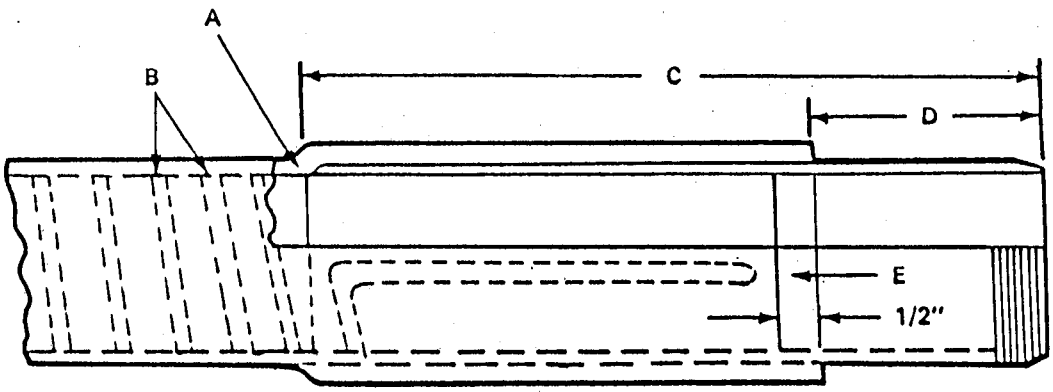
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1A. Hose ends for banded shank, compression, or internally expanded fittings.



1B. Hose ends for built-in nipples.



1C. Hose ends for plain nipples.

A. Rubber and reinforcement plies.
B. Helix wire.
C. Overall nipple length (see figures 5 and 6)
D. Length of nipple beyond hose end (see table I).
E. Soft rubber end.

FIGURE 1. Hose end constructions.

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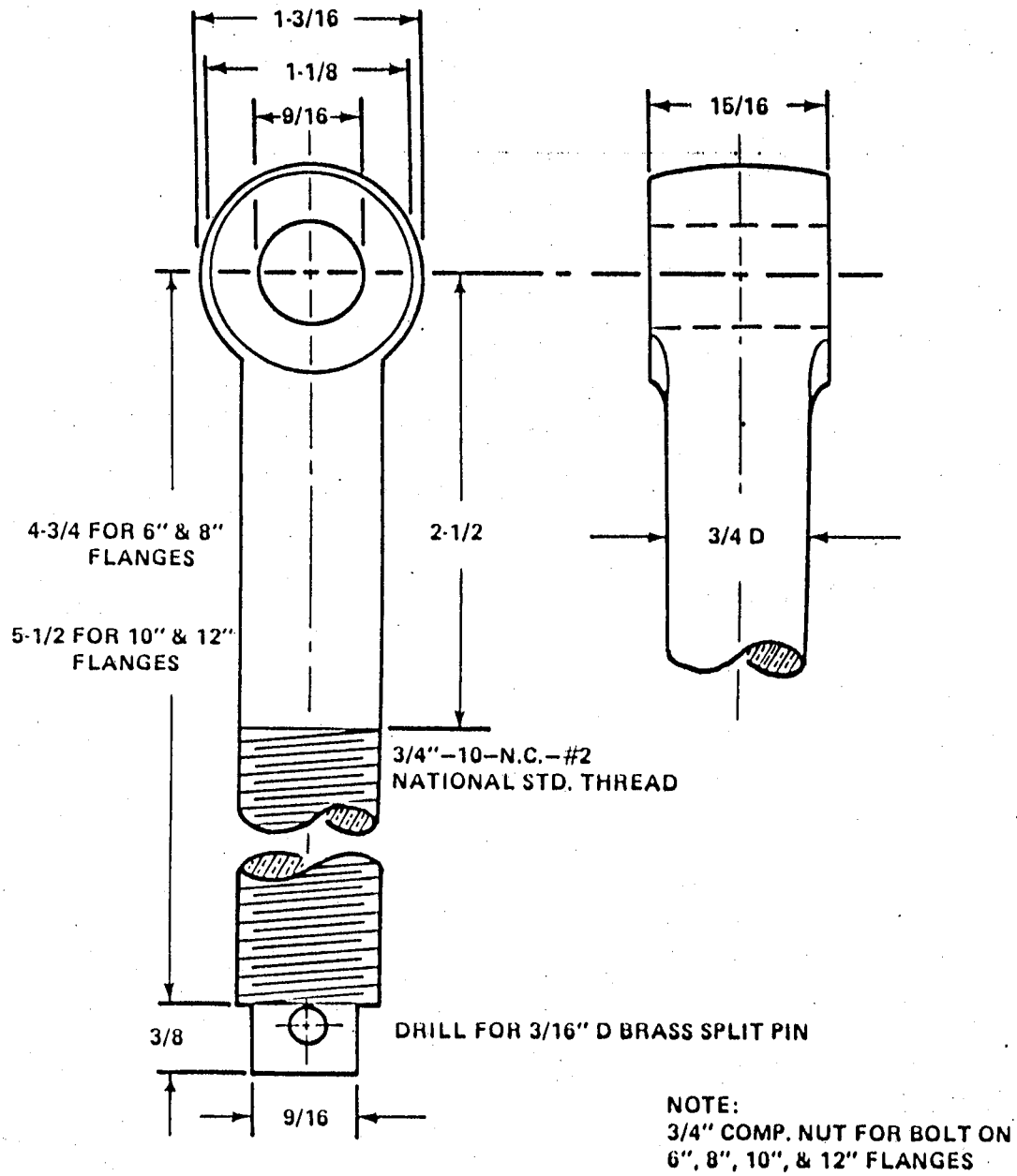
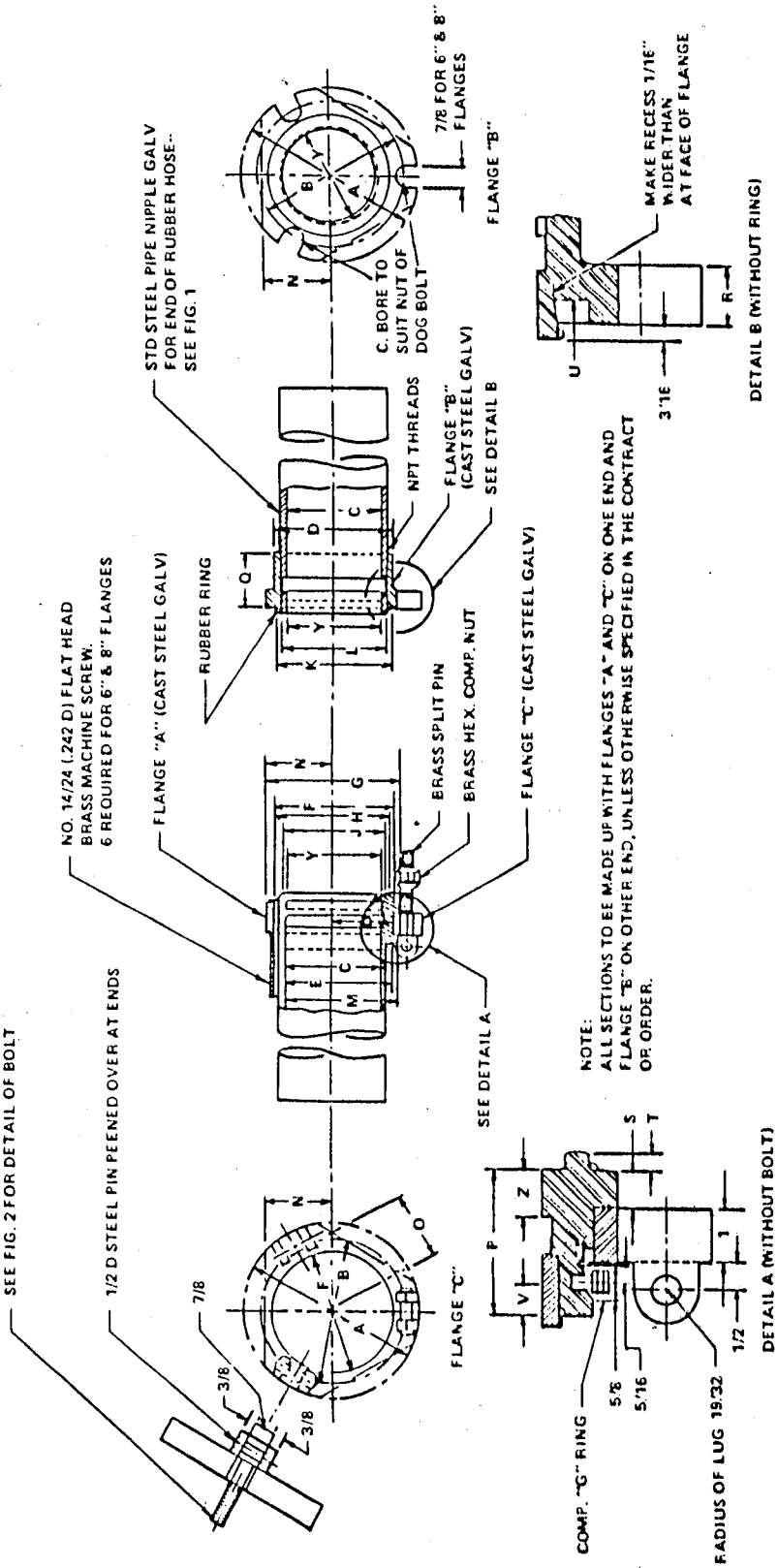


FIGURE 2. Bolt for flange.

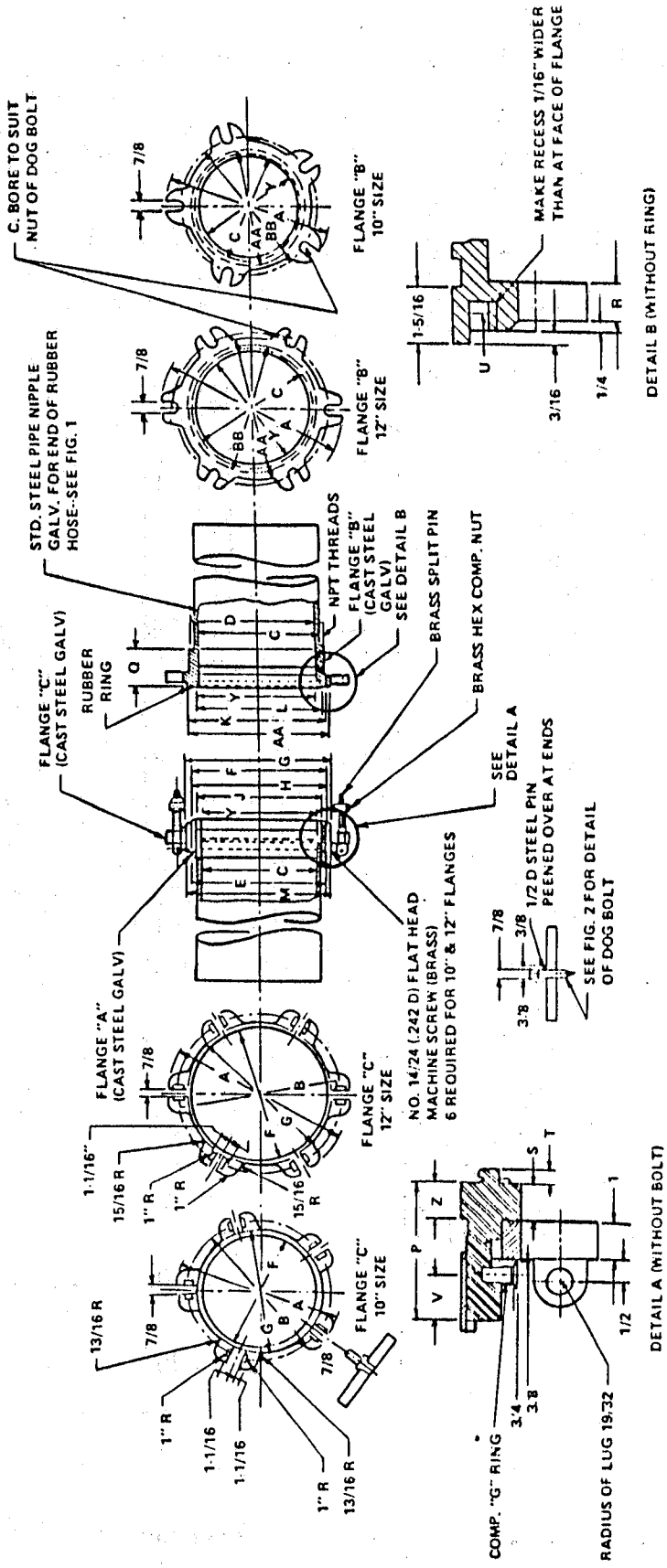
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FLANGE SIZE (INCHES)												
	6	8		6	8		6	8		6	8
A	11	13	J	6 9/16	8 5/8	R	1 3/16	7/8			
B	9 1/4	11 1/4	K	7 1/2	9 1/4	S	7/16	9/16			
C	6 5/8	8 5/8	L	6 1/2	8 1/2	T	5/16	5/16			
D	7 3/8	9 3/8	M	7 7/8	7 7/8	U	5/16	3/8			
E	7 3/8	9 3/8	N	4 1/4	4 1/4	V	11/16	7/8			
F	7 1/2	9 1/2	O	4 3/16	5 3/16	Y	6	3/4			
G	8 3/8	10 3/8	P	2 1/2	2 13/16	Z	8	3/4			
H	7 7/16	9 1/2	Q	2 5/8	3						

FIGURE 3. Flanges 6- and 8-inch.

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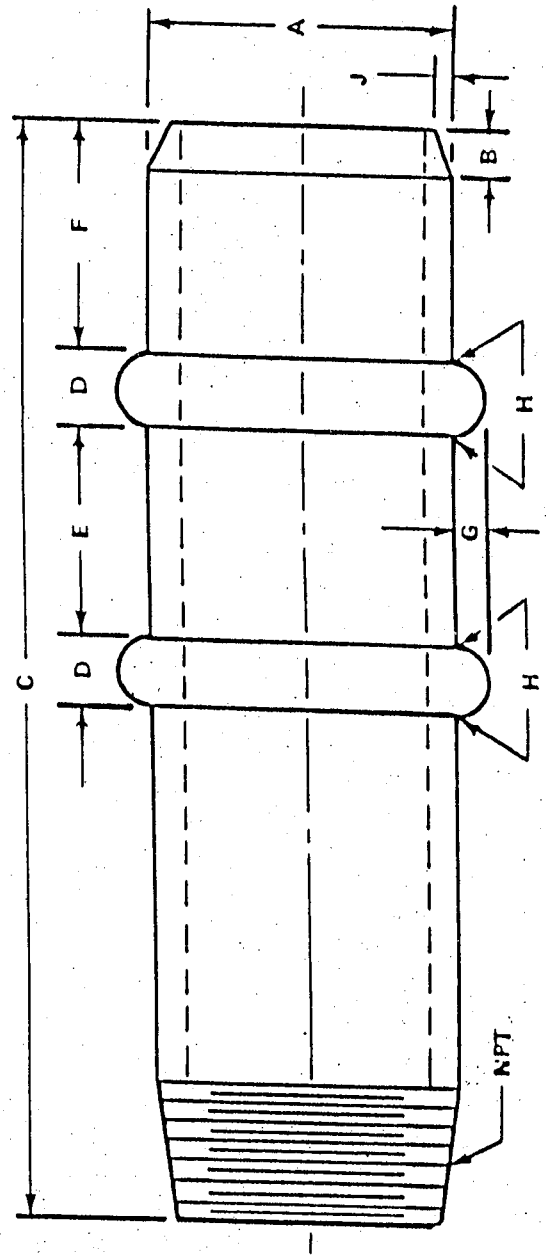


NOTE: ALL SECTIONS TO BE MADE UP WITH FLANGES "A" AND "C" ON ONE END AND FLANGES "B" ON OTHER END, UNLESS OTHERWISE SPECIFIED IN THE CONTRACT ORDER.

FLANGE SIZE (INCHES)												
10		12		10		12		10		12		
A	16	3/4	18	3/4	J	11	1/4	12	7/8	T	5/16	5/16
B	14	3/4	16	3/4	K	11	13/16	13	15/16	U	3/8	3/8
C	10	3/4	12	3/4	L	11	3/16	12	11/16	V	1 3/16	1 3/16
D	11	3/4	13	3/4	M	12	3/8	14	3/8	Y	10	12
E	11	3/4	13	3/4	P	3	1/2	3	1/2	Z	7/8	7/8
F	11	7/8	13	7/8	Q	3	3/4	3	3/4	AA	12 7/8	14 7/8
G	12	7/8	14	7/8	R	1		1		BB	14 1/8	16 1/8
H	11	3/4	13	3/4	S		7/8		7/8			

FIGURE 4. Flange 10- and 12-inch.

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Nipple dimensions in inches

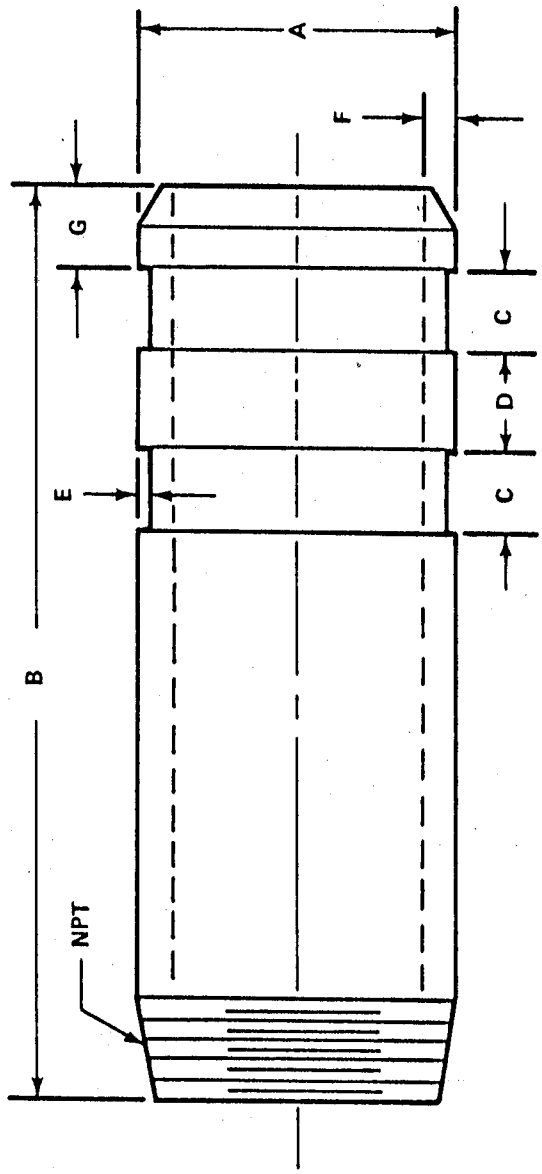
Size	A*	B	C	D	E	F	G	H	J
		±0.0625	±0.0625	±0.09375	±0.1875	±0.1875	±0.0625	±0.0625	±0.03125
4	4.500	0.500	16.0	0.75	4	4	0.375	0.1875	0.125
6	6.625	0.500	21.0	1.00	5	5	0.500	0.1875	0.125
8	8.625	0.500	21.0	1.00	5	5	0.500	0.1875	0.125
10	10.750	0.500	23.5	1.25	6	5	0.625	0.250	0.125
12	12.750	0.500	24.0	1.50	6	5	0.750	0.315	0.125

*Dimensions conform to ASTM A53, schedule 40.

Note: Nipple bands may have square shoulders.

FIGURE 5. Nipple for Style C hose.

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Nipple dimensions in inches

Size	A*	B	C	D	E	F*	G	Groove
		Min	±0.0625	±0.0625	±0.03125	±0.03125	±0.0625	Min
4	4.500	15	1.50	1.50	0.0625	0.337	1.50	2
6	6.625	16	2.00	2.00	0.1875	0.432	1.50	2
8	8.625	18	2.00	2.00	0.1875	0.500	1.50	2
10	10.750	20	2.00	2.00	0.2500	0.500	2.00	2
12	12.750	20	2.00	2.00	0.2500	0.500	2.00	2

*Dimensions conform to ASTM A53.

FIGURE 6. Nipple for Style D hose.

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